

Derivation and Numerical Reproduction of Liang, Goldsman, Mayergoyz & Oldiges (1997): 2-D Deterministic SHE-Boltzmann MOSFET Simulation

*Complete report, Revision 4 — reduced P1/SHE-Boltzmann reproduction of Liang’s 2-D
signatures; audit-response revision*

Reproduction study (code commit 0994290+overlays; Python 3.13.14, NumPy 2.5.1, SciPy 1.18.0)

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Abstract

This report derives and numerically reproduces the deterministic spherical-harmonic (SHE) Boltzmann device solver of Liang *et al.*, *IEEE Trans. Electron Devices* **44**(2), 257–267 (1997). Revision 2 superseded the earlier one-way/post-processed description: impact ionization is now *inside* the assembled electron collision operator (primary depletion + secondary redistribution), the self-consistent SHE $\leftrightarrow$ Poisson $\leftrightarrow$ hole-continuity loop is *demonstrated and observably converging*, global number- and terminal-current-conservation diagnostics are in place, and a physics-split preconditioner removes the previous solver bottleneck (~ 10 min $\rightarrow \sim 15$ s on the 40×34 problem). Established results: core transport (n , $|v|$, T_e) is validated and spatially robust; the self-consistent feedback materially suppresses avalanche generation and drain current while leaving T_e comparatively stable; conservation holds across one-way, refined, and coupled runs. Open items are stated precisely: the impact-ionization magnitude is *not spatially converged* — a 2×2 grid $\times$ coupling study drives $G_{ii,\max}$ multiplicatively from 1.35×10^{27} to 5.24×10^{26} cm $^{-3}$ s $^{-1}$ (the finest-tested coupled value, not a continuum estimate), while a one-way energy refinement to $\Delta H = 6.25$ meV moves it only -2.4% (negligible beside the spatial error); the T_e/G_{ii} peak offset is unresolved; and the device has been recalibrated to $V_{th} = 0.614$ V for the quantitative Liang overlays. The robust reproductions are the transport moments (n , $|v|$). **A 50-iteration convergence study (§9) shows the self-consistent G_{ii} does *not* converge in the outer coupling loop** — it decreases monotonically with no plateau ($2.1 \times 10^{27} \rightarrow 6.3 \times 10^{26}$, still falling as $\max|\Delta\phi|$ reaches 2.5 mV, short of the 2 mV criterion) — so no self-consistent G_{ii} magnitude is quoted, and the previously-reported -37% feedback fraction is superseded by an ongoing suppression of at least -53% . The quantitative overlays (§10) use the best-converged (50-iteration) fields on Liang’s *own* published Fig. 7 contour levels: $T_{e,\max} \approx 3323$ K (66% of Liang’s top 5000 K contour) and $G_{ii,\max} \approx 6.3 \times 10^{26}$ (27% of the top 2.3×10^{27} , itself unconverged), both drain-localized with the correct structure (T_e region much broader than G_{ii}); a one-way *SHE velocity-moment* I-V reproduces Liang’s Fig. 8 method — terminal current from the SHE face flux, not a mobility model (source/drain conservation $\leq 10^{-3}$), running $0.5\text{--}0.96\times$ the DD proxy — and a per-plot comparison table is given for each of Liang’s Figs. 2–8. **Revision 4** (this version) responds to an external audit by reframing the work honestly: it is a *reduced* P1/($\ell \leq 1$) SHE-Boltzmann reproduction of the principal 2-D transport and impact-ionization *signatures*, using reconstructed device data and *surrogate scattering and band models*. Liang’s explicit surface-roughness, surface-acoustic, and ionized-impurity operators are *lumped* into a single calibrated momentum time τ_1 , not reproduced as distinct mechanisms (Table 2); the quantitative overlays use a *calibrated* device (V_{th} fitted to 0.614 V), so they are calibrated comparisons, not blind predictions (Table 3); and the DD I-V curve is retained only as a comparison proxy, while the one-way SHE face-flux sweep reproduces Liang’s current-extraction method (§10). Every model element is classified as *paper-specified*, *reconstruction*, *surrogate/lumped*, or *unconverged/mesh-dependent*.

Contents

1	Overview and audit provenance	2
I	Method derivation and validated foundation	2
2	First-order SHE equation	3
3	Reconstructed device, taxonomy, foundation	3
II	The 2-D SHE core, impact ionization, and self-consistency	3
4	Assembled operator, impact ionization, and moments	4
5	Numerics: physics-split preconditioner	5
6	Numerical audit of the one-way baseline	5
7	Grid refinement (40×34 vs 60×50 , one-way)	5
8	Self-consistent $\text{SHE} \leftrightarrow \text{Poisson} \leftrightarrow \text{hole}$ loop	6
9	G_{ii} convergence matrix and device recalibration	6
10	Quantitative comparison with Liang’s published figures	8
III	Assessment	12
11	Conclusions, classified	13
A	Reproducibility package	15
B	Provenance and audit history	16

1 Overview and audit provenance

Liang *et al.* solve, self-consistently, three coupled equations on a 2-D MOSFET cross-section: Poisson, the electron BTE (first-order SHE), and hole continuity with an impact-ionization pair-generation source. This reproduction has passed through four external audit rounds and three code patches (Appendix B); the present revision documents the resulting coupled state and its quantitative comparison against Liang’s figures.

Honesty note. Model substitutions carried throughout and classified in Table 1: the doping profile is a *reconstruction*; the band is a 1-parameter Kane *surrogate* (paper: 2-band Fiegna/Brunetti); the impact-ionization rate is a Keldysh *surrogate*; the momentum time τ_1 is a single calibrated surrogate (not a summed Table-I model); and the device V_{th} , initially an over-high 1.64 V, is recalibrated to 0.614 V (§9) for the quantitative overlays (§10).

Part I

Method derivation and validated foundation

2 First-order SHE equation

With $\mathbf{E} = -\nabla_{\mathbf{r}}\phi$, expand $f = f_0(\mathbf{r}, \varepsilon) + \hat{\mathbf{\Omega}} \cdot \mathbf{f}_1(\mathbf{r}, \varepsilon)$ (all $\ell \geq 2$ discarded). This $\ell \leq 1$ (P1) truncation is Liang’s “first four spherical harmonics”: Y_0^0 plus the three $\ell = 1$ components ($Y_1^{-1,0,1}$) carried by $\hat{\mathbf{\Omega}} \cdot \mathbf{f}_1$. Projection gives the current relation and the conservative augmented-space balance

$$\mathbf{f}_1 = -\tau_1 v (\nabla_{\mathbf{r}} f_0 - q \mathbf{E} \partial_{\varepsilon} f_0), \quad \nabla_{\mathbf{r}} \cdot \mathbf{j}_x + \partial_{\varepsilon} j_{\varepsilon} = Z \mathcal{Q}[f_0] + Z S_{\text{gen}}, \quad (1)$$

$\mathbf{j}_x = -ZD(\nabla_{\mathbf{r}} f_0 - q \mathbf{E} \partial_{\varepsilon} f_0)$, $D = \frac{1}{3}v^2\tau_1$. In total energy $H = \varepsilon - q\phi$ this becomes pure spatial diffusion at fixed H , the collision operator coupling $H \leftrightarrow H \pm E_{op}$ locally in space, over the global mesh $-q\phi_{\max} \leq H \leq \varepsilon_{\max} - q\phi_{\min}$ with mask $0 \leq H + q\phi \leq \varepsilon_{\max}$; $\varepsilon = 0$ reflecting, $\varepsilon_{\max}$ absorbing. The DOS-weighted acoustic operator $\mathcal{Q}_{ac} = Z^{-1}\partial_{\varepsilon}[ZA(\partial_{\varepsilon} + 1/k_B T)f_0]$ (Scharfetter–Gummel) makes the collision term irreducible; it is number-conserving (defect 7.6×10^{-17}) with the Maxwellian as its exact null.

3 Reconstructed device, taxonomy, foundation

The 0.35 μm LDD NMOS uses error-function junctions ($x_j = 0.170 \mu\text{m}$ metallurgical). Foundation checks (all at $T = 300 \text{ K}$): equilibrium Poisson (built-in $+0.586 / -0.44 \text{ V}$, quadratic Newton); nonparabolic $N_c = 2.857 \times 10^{19}$ (0.1% of the independent value; parabolic limit 0.09% of closed form); bulk scattering engine detailed-balance 2×10^{-5} , zero-field $T_e \rightarrow 309.7 \text{ K}$, monotone heating. The DD solve at $V_{gs} = V_{ds} = 3 \text{ V}$ gives the operating-point potential driving the SHE.

Table 1: Modelling taxonomy (Rev. 2 adds a *not-yet-converged* class).

Element	Class	Basis / status
Method eqs., SHE reduction, moments	paper	Liang <i>et al.</i>
Table-I params, spherical band	paper	paper Sec. II
$L_{\text{eff}}, t_{ox}, x_j, N_D^{\pm}$	paper	paper text
1-param Kane $\gamma(\varepsilon)$	surrogate	paper: 2-band Fiegna/Brunetti
LDD doping shape, mesh	reconstruction	ref. [25] (Khalil 1994) unavailable
Surface-roughness / -acoustic / impurity scattering	lumped surrogate	folded into τ_1 ; not distinct operators (Table 2)
Channel $N_A, \Phi_{gate}, \tau_1, c_{op}, 1/\tau_{ii}$	surrogate	calibrated; disclosed (Table 3)
$G_{ii, \max}$ magnitude	unconverged	mesh-dependent; 5.24×10^{26} (coupled 60×50), still dropping
T_e/G_{ii} peak offset	unconverged	unresolved at present mesh

Table 2: Paper-to-code physics matrix: every Liang scattering mechanism, and how it is treated here. “Lumped” means folded into the single calibrated momentum time τ_1 rather than implemented as a distinct spherical-harmonic collision operator.

Mechanism	Liang <i>et al.</i>	This work	Status
Acoustic phonon (bulk)	explicit elastic op.	DOS-weighted SG operator	present (reduced)
Optical / intervalley phonon	explicit inelastic op.	detailed-balance operator, surrogate c_{op}	present (surrogate)
Ionized impurity (Brooks–Herring)	explicit $\ell=1$ op.	folded into τ_1	lumped
Surface acoustic phonon (D_{sa})	explicit op.	folded into τ_1	lumped
Surface roughness (Γ, δ)	explicit op.	folded into τ_1	lumped
Impact ionization	numerical rate [23]	Keldysh surrogate, in operator	present (surrogate rate)
e – h recombination (SRH)	in BTE (trap occupancy)	SRH in DD hole continuity	proxy location
Net momentum relaxation	sum of the above	single calibrated τ_1	lumped surrogate

Table 3: Calibration vs. validation. Fitted parameters, the target they were fitted to, the fitting data, and an *independent* quantity used to check them. The quantitative overlays are therefore calibrated comparisons, not blind predictions.

Parameter	Fitted target	Fitting data	Independent quantity	comparison
$\Phi_{gate} = -0.74$ V	$V_{th} = 0.614$ V	extracted threshold	I_d – V_g	subthreshold swing (~ 99 mV/dec); overlay
$N_{A, \text{chimp}} = 4 \times 10^{17}$	V_{th}	threshold extraction		hot-spot location
$\tau_1 = 10^{-14} / \sqrt{\gamma \gamma'}$	bulk mobility / $ v _{sat}$	bulk $ v $ – E curve	n_{surf} device $ v _{max} \approx 10^7$ cm/s	surface inversion density
$c_{op} = 4.69 \times 10^{-34}$	bulk $T_e(E)$ heating	bulk hot-electron T_e	device $T_{e, max}$ localization	
$1/\tau_{ii} (P=2 \times 10^{13}, \varepsilon_{th}=1.1$ eV)	II onset threshold	Keldysh-type surrogate	G_{ii} drain localization; feedback fraction	

Part II

The 2-D SHE core, impact ionization, and self-consistency

4 Assembled operator, impact ionization, and moments

The solver discretizes the $(\mathbf{r}, H)$ equation by box integration; contributions (spatial diffusion $\times \Delta H$, optical in/out $\times A_{ij} \Delta H$, acoustic SG flux $\times A_{ij}$) share units and form a positive-diagonal system. Contacts inject a Maxwellian $f_0 = n_{contact} e^{-\varepsilon/k_B T} / \int Z e^{-\varepsilon/k_B T} dH$.

Impact ionization in the electron BTE. An event removes a primary electron at rate $1/\tau_{ii}(\varepsilon)$ (Keldysh surrogate, $\varepsilon_{th} = 1.1$ eV) and deposits two secondaries at $\varepsilon' = (\varepsilon - \varepsilon_{th})/2$, linearly distributed onto the neighbouring H nodes, with net electron gain +1 per event. This is *assembled into the electron collision operator* (primary depletion + secondary redistribution), so f_0 is directly shaped by avalanche; $G_{ii} = \int Z f_0 / \tau_{ii} d\varepsilon$ is a moment of that II-affected distribu-

tion. The equal kinetic-energy split of the two secondaries, and the fact that this scattering-in term leaves the f_0 shape essentially unchanged, follow Liang *et al.* (§IV) directly. The absorbing ε_{max} cutoff acts on spatial, acoustic, and optical fluxes. Moments n , $\langle \mathbf{v} \rangle$, T_e , G_{ii} , and the terminal currents are reconstructed from the *same finite-volume face fluxes* as the operator.

5 Numerics: physics-split preconditioner

The large-grid system is solved by a multiplicative physics-split preconditioner. Because DOFs are numbered with H fastest, same-spatial-cell couplings (acoustic + optical + II redistribution + spatial diagonal) form contiguous local energy blocks — the **E block** — factored cheaply; the diagonal plus cross-cell couplings form the fixed- H real-space **S block**. One application does $E^{-1} \rightarrow S^{-1} \rightarrow E^{-1}$ with a fresh residual between stages; the solved matrix is never shifted (shifts touch only preconditioner factors). On the 328 544-DOF 40×34 problem this solves to 10^{-10} in ~ 15 s (raw scaled residual 1.22×10^{-11}), where the previous global ILU did not finish its setup in 120 s. The legacy path is retained as `method="legacy_ilu"` for regression.

6 Numerical audit of the one-way baseline

Table 4 is the extracted audit of the locked one-way reference (commit `f016e3c`); it is preserved as a regression benchmark.

Table 4: Clean 40×34 numerical audit (impact ionization in the collision operator).

Quantity	Value	Location / note
$n_{\max}$	$1.0516 \times 10^{20} \text{ cm}^{-3}$	source n^+ (0.082, 0.043)
$T_{e,\max}$	3232.9 K	drain (0.595, 0.005)
$G_{ii,\max}$	$1.3513 \times 10^{27} \text{ cm}^{-3} \text{ s}^{-1}$	drain (0.595, 0.009)
$ v _{\max}$	$9.732 \times 10^6 \text{ cm/s}$	drain surface (0.554, 0)
$T_e - G_{ii}$ peak separation	5 nm (~ 1 cell)	co-located within mesh
SHE currents I_{src}/I_{drn}	$-1.7044/ + 1.7071 \times 10^{-4} \text{ A}/\mu\text{m}$	mismatch 0.16%
global continuity residual	-5.70×10^{-6} (relative)	—
orphan states	0	no artificial sinks
f_0 pre-clip min	-3.3×10^{-11}	46/328544 tiny negatives

7 Grid refinement (40×34 vs 60×50 , one-way)

Refining the spatial mesh at fixed $\Delta H = 12.5 \text{ meV}$:

Quantity	40×34	60×50	converged?
$n_{\max}$	1.0516×10^{20}	1.0473×10^{20}	yes (-0.4%)
$ v _{\max}$	9.73×10^6	10.0×10^6	yes ($+2.7\%$)
$T_{e,\max}$ [K]	3232.9	3423.0	moderate ($+5.9\%$)
$G_{ii,\max}$	1.3513×10^{27}	8.26×10^{26}	no (-40%)
T_e/G_{ii} separation	1 cell	2 cells	grid-scale (unresolved)
current mismatch / residual	0.16% / 5.7×10^{-6}	0.07% / 7.6×10^{-6}	conserved

n and $|v|$ are spatially robust over this refinement; T_e is moderately grid-sensitive (5.9%, not converged); G_{ii} is *not* grid-converged (it depends on the high-energy tail); and the peak separation tracks the mesh, so it is unresolved, not a reproduced physical offset.

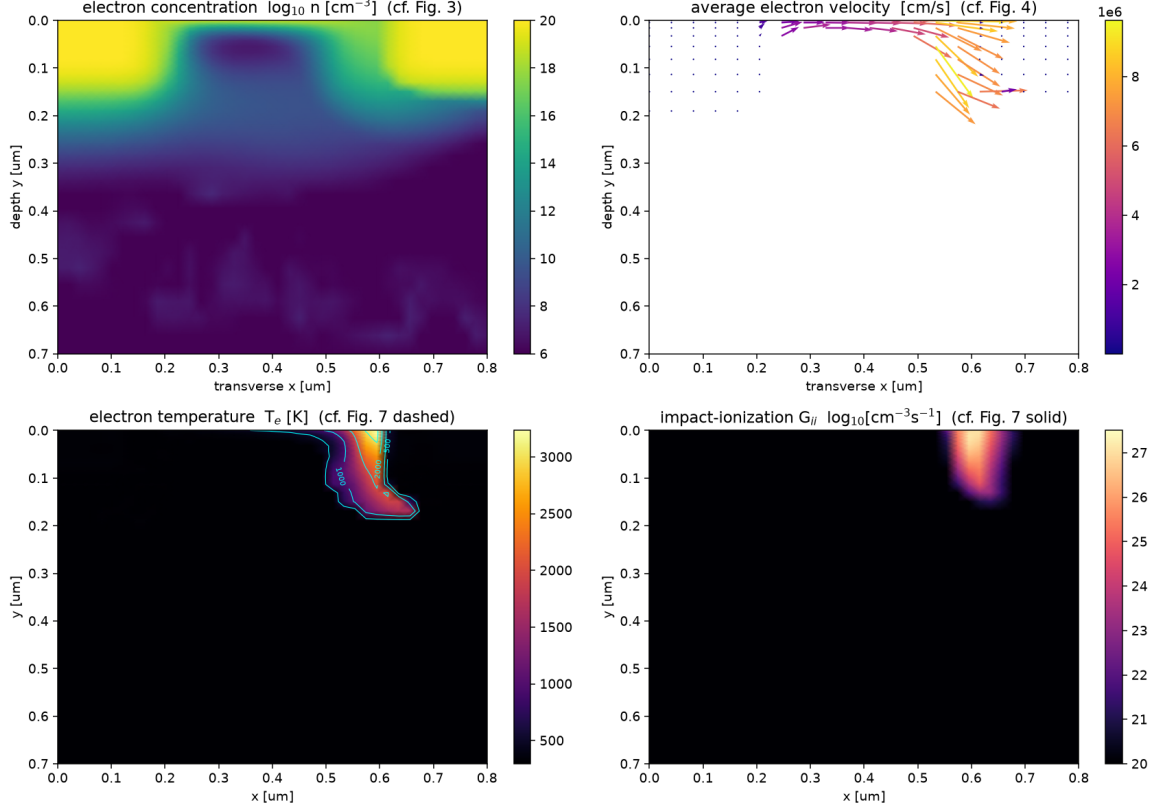


Figure 1: One-way SHE moments (40×34). n (Fig. 3), velocity field saturating at 9.73×10^6 cm/s (Fig. 4; masked where $n < 10^{13}$ cm $^{-3}$), T_e (Fig. 7 dashed, drain-localized), G_{ii} (Fig. 7 solid, drain-localized).

8 Self-consistent SHE $\leftrightarrow$ Poisson $\leftrightarrow$ hole loop

A damped outer iteration solves the SHE (with impact ionization) on the current potential, drives hole continuity with G_{ii} as pair generation (holding n_{SHE}), updates Poisson from the SHE density, damps, and repeats; n is recomputed from the SHE after each potential update (not constitutively from DD). Over 12 iterations (Fig. 2) the maximum potential update falls monotonically $17.7 \rightarrow 4.0$ mV, and over these first 12 iterations the observables *appeared* to plateau: $G_{ii,\text{max}}$ suppressed 37% ($1.35 \rightarrow 0.85 \times 10^{27}$), drain current 29% ($1.71 \rightarrow 1.21 \times 10^{-4}$ A/ μm), $T_{e,\text{max}} \sim 3240$ K. **This early plateau is only apparent:** the extended 50-iteration study (§9, Fig. 3) shows G_{ii} continues to fall monotonically. This is the physically expected negative feedback: the SHE electrons and the G_{ii} -generated holes reshape the potential, lowering the peak drain field and throttling avalanche. Final conservation is clean (relative number balance 2.4×10^{-5} , currents balance 0.14%).

Honesty note. This is a 12-iteration *snapshot*. The loop is not converged ($\max |\Delta\phi| \approx 4$ mV, above the 2 mV criterion), and the apparent G_{ii} plateau does *not* survive: extending to 50 iterations (§9) shows G_{ii} keeps decreasing. n and $|v|$ are the robust observables.

9 G_{ii} convergence matrix and device recalibration

Combining one-way and coupled runs at both grids isolates how the impact-ionization magnitude responds to spatial refinement and to self-consistency (Table 5). The spatial ratios ($8.26/13.5 = 0.612$ one-way, $5.24/8.50 = 0.616$ coupled) and the coupling ratios ($8.50/13.5 = 0.630$ at 40×34 ,

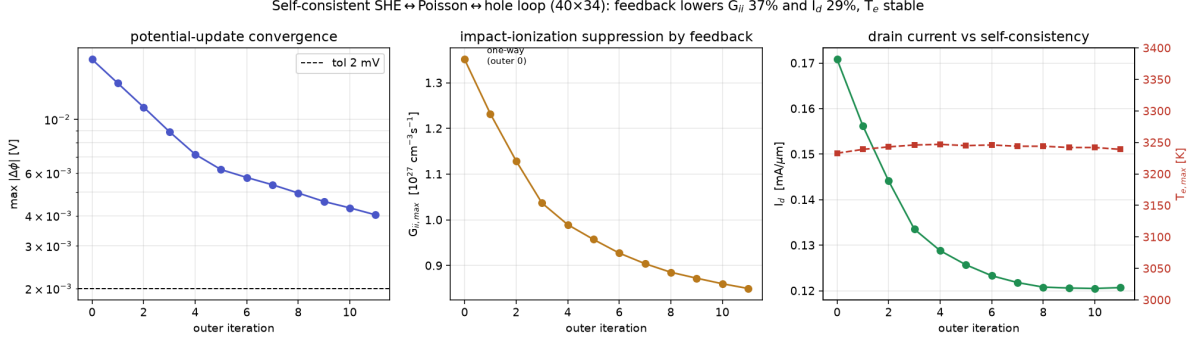


Figure 2: Self-consistent loop (40×34), first 12 iterations. Potential update $17.7 \rightarrow 4.0$ mV; feedback suppresses G_{ii} and I_d with an *apparent* early plateau — superseded by the 50-iteration study (§9, Fig. 3), where G_{ii} keeps falling.

$5.24/8.26 = 0.634$ at 60×50) agree to $\sim 1\%$ (ratio-of-ratios 1.0085): the two effects are *multiplicatively separable to that precision* over the tested meshes, and $1.35 \times 10^{27} \times 0.612 \times 0.630 = 5.20 \times 10^{26}$ predicts the observed 5.24×10^{26} . This is an *empirical relation among the matched coupled-snapshot (under-converged) calculations*, not a converged self-consistent scaling law — the coupled entries are not fixed-point values (Fig. 3). The coupled result is *still* spatially unconverged (coupled $40 \times 34 \rightarrow 60 \times 50$: -38% , no plateau), and, like the 40×34 loop, the coupled 60×50 outer iteration ended above the 2 mV $\max |\Delta\phi|$ criterion. As the 50-iteration recalibrated study below shows (Fig. 3), the coupled loop does *not* converge G_{ii} at all — it decreases monotonically — so the self-consistent column here holds under-converged snapshots, not fixed-point values.

Table 5: $G_{ii,\max}$ [$\text{cm}^{-3}\text{s}^{-1}$] over grid $\times$ coupling (original device). The coupled column holds *under-converged snapshots*: the loop does not converge G_{ii} (§9, Fig. 3).

	one-way	coupled (snapshot)
40×34	1.35×10^{27}	8.50×10^{26}
60×50	8.26×10^{26}	5.24×10^{26}

A *one-way* energy refinement to $\Delta H = 6.25$ meV ($E_{op} = 8$) moves $G_{ii,\max}$ only -2.4% ($T_e - 0.19\%$) — *negligible beside* the -38 – 40% spatial error — so the dominant numerical uncertainty is resolving the drain-side high-field/high-energy hotspot in real space rather than the energy axis. (The energy test was performed one-way, not on the coupled configuration in which the spatial sensitivity was measured, so we state energy insensitivity, not a proof of purely-spatial error.) 5.24×10^{26} is thus the *finest-tested coupled value*, not an estimate of the continuum limit: with a -38% change under the last spatial refinement there is no numerical basis to quote a converged G_{ii} . The robust results are the transport moments (n , $|v|$); the feedback suppression is *not* converged — it reaches at least -53% (§9 below) and keeps growing.

Device recalibration and coupled-loop convergence study. A rigid gate offset $\Phi_{gate} = -0.74$ V ($dV_{th}/d\Phi_{gate} \approx +1$) recalibrates the extracted threshold to $V_{th} = 0.614$ V (was 1.64 V); the $V_g = 3$ V overdrive becomes 2.39 V. A 12-outer-iteration run (stopped at $\max |\Delta\phi| = 7.5$ mV) gave a snapshot $G_{ii,\max} = 1.34 \times 10^{27}$; **extending the loop to 50 iterations (Fig. 3) shows that snapshot was under-converged.** Over 50 damped iterations (ϕ -damping 0.35) $\max |\Delta\phi|$ falls $17.1 \rightarrow 2.5$ mV — *not* reaching the 2 mV criterion — and the observables split sharply by convergence behaviour: $n_{\max} = 1.02 \times 10^{20}$ (last-4 relative spread 0.2%) and $|v|_{\max} = 9.9 \times 10^6$ cm/s are *stable*; $T_{e,\max}$ drifts down to 3323 K; $I_d \approx 0.28$ mA/ μm ; but $G_{ii,\max}$ *decreases*

monotonically with no plateau, $2.1 \times 10^{27} \rightarrow 6.3 \times 10^{26}$ (a factor 0.30), and is *still falling* at iteration 50. **So the self-consistent G_{ii} is not converged in the outer coupling loop either** — not only on the mesh. The earlier 1.34×10^{27} was a snapshot; the feedback suppression is at least -53% (vs. the previously-quoted -37%) and still growing; and *no self-consistent G_{ii} magnitude can be quoted*. The overlays of §10 use the best-converged (50-iteration) fields, on which T_e reaches 66% and G_{ii} 27% of Liang’s top contours — but the G_{ii} fraction is itself unconverged and decreasing.

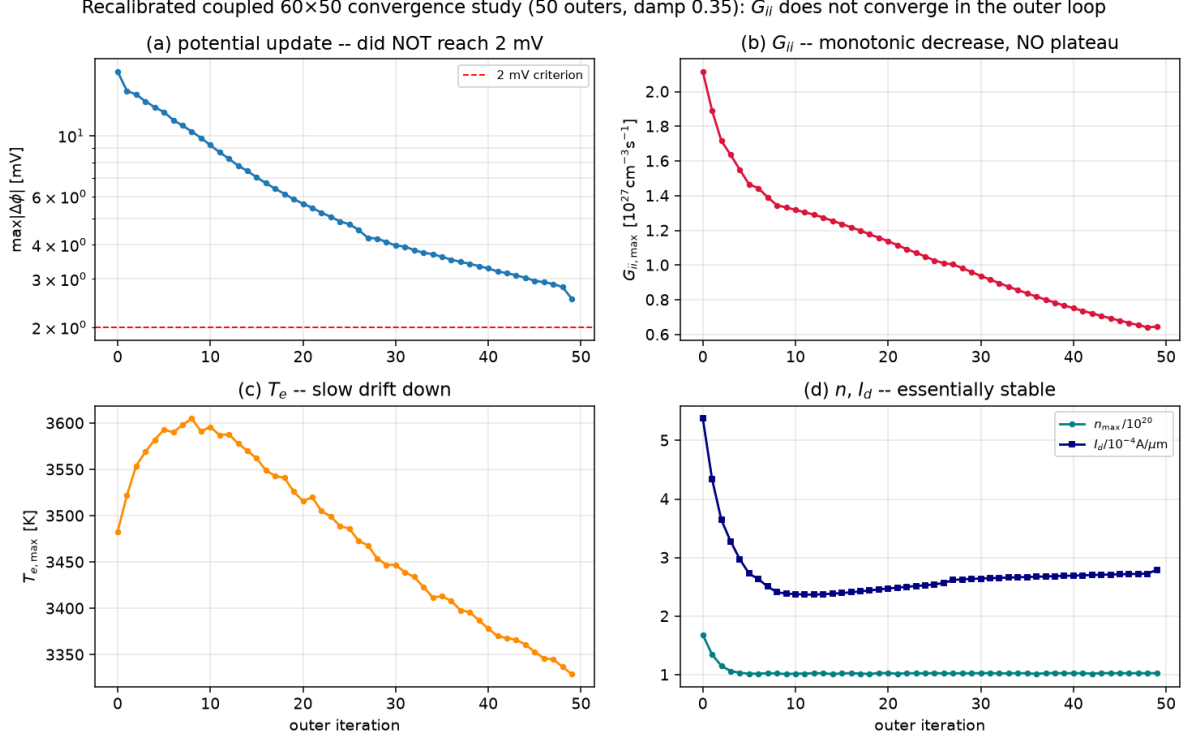


Figure 3: Recalibrated coupled 60×50 convergence study (50 outer iterations, ϕ -damping 0.35). (a) $\max|\Delta\phi|$ falls to 2.5 mV, short of the 2 mV criterion; (b) $G_{ii,\max}$ decreases monotonically with *no plateau* — not converged in the outer loop; (c) T_e drifts slowly downward; (d) n and I_d are essentially stable.

10 Quantitative comparison with Liang’s published figures

The recalibrated coupled 60×50 solution (§9, $V_{th} = 0.614$ V) is compared to Liang *et al.*’s figures below, with *one comparison table per plot* (Figs. 2–8, Tables 6–12). Where Liang give labelled contour values (Fig. 7) the comparison is on those values; otherwise it is on peak magnitudes, locations, and structure. Liang’s Fig. 7 levels (paper caption, p. 263) are $T_e \in \{500, 1000, 2000, 3000, 4500, 4800, 5000\}$ K (dashed) and $G_{ii} \in \{1.0, 4.0\} \times 10^{26}, \{1.0, 1.7, 2.3\} \times 10^{27} \text{ cm}^{-3}\text{s}^{-1}$ (solid). (A Rev. 3 transcription that omitted the 4800 K level and added a spurious $2.3 \times 10^{26} G_{ii}$ level was corrected in Rev. 4, and the overlays regenerated.)

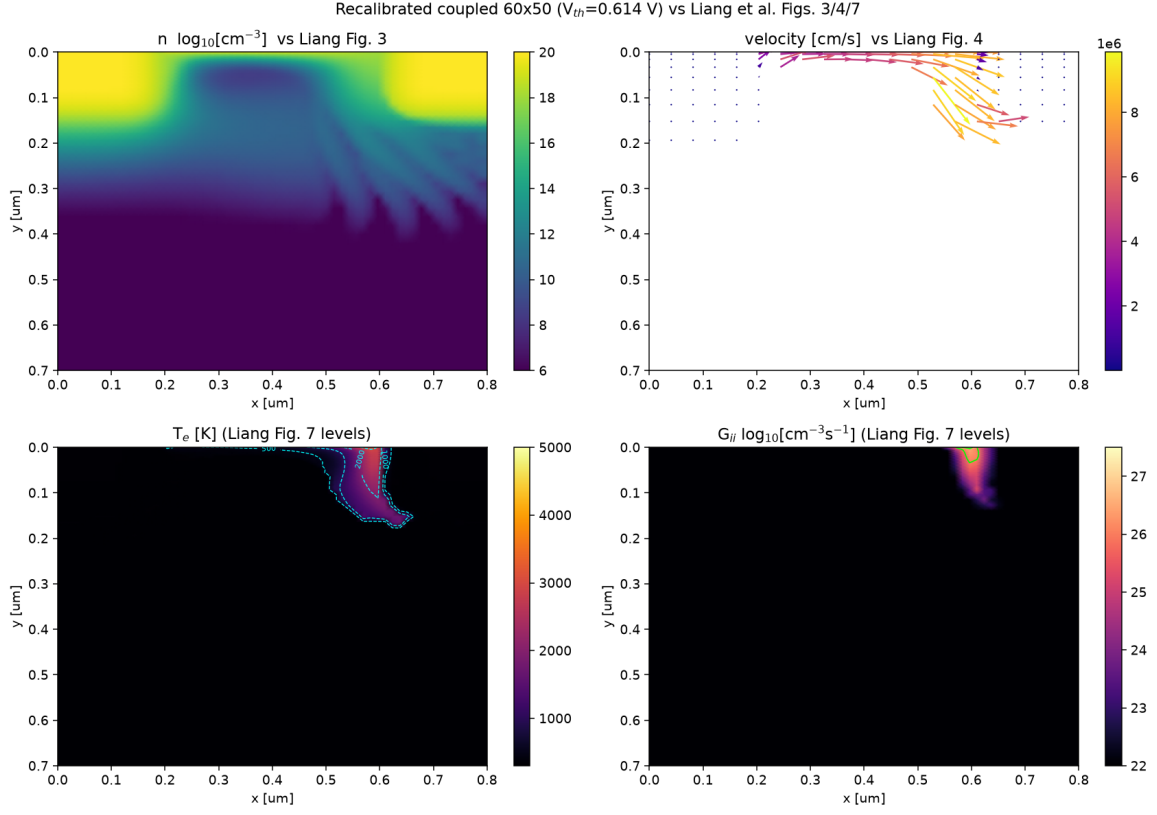


Figure 4: This work vs Liang Figs. 3 (n), 4 (velocity), 7 (T_e , G_{ii} at Liang's contour levels).

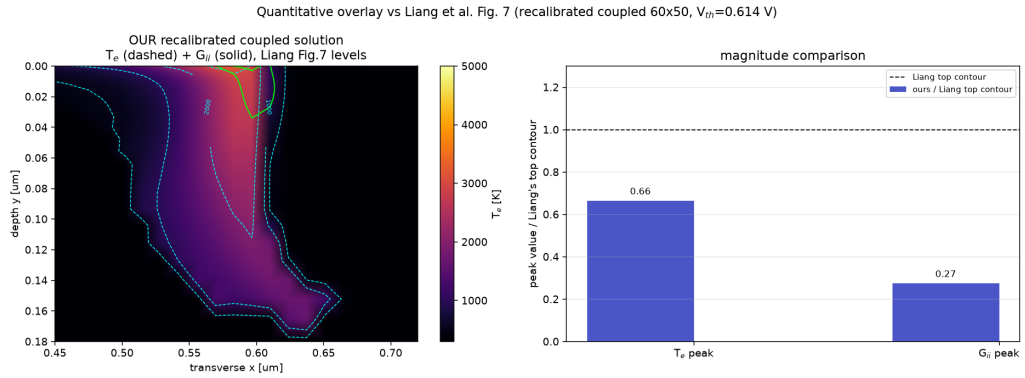


Figure 5: Quantitative Liang Fig. 7 overlay: our T_e (dashed) and G_{ii} (solid) on Liang's exact published levels; peaks reach 0.66 (T_e) and 0.27 (G_{ii} , unconverged) of the top contours.

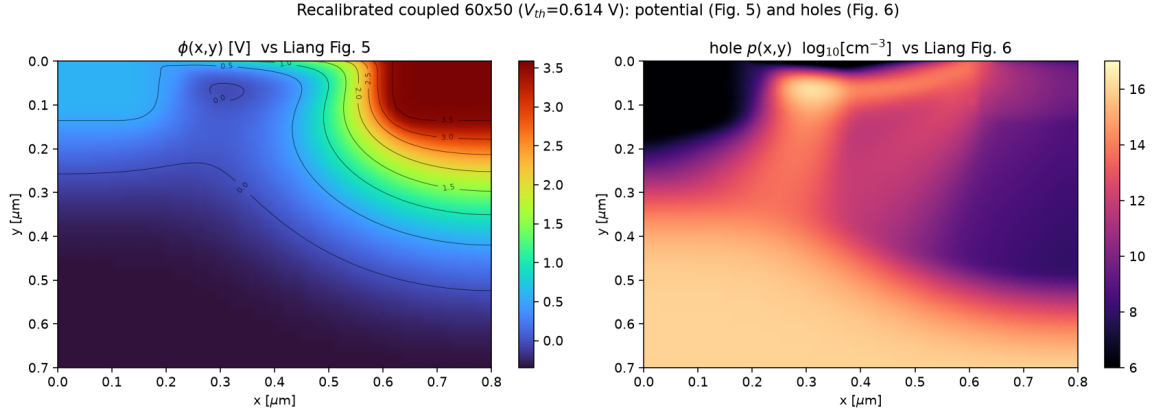


Figure 6: Potential ϕ (Liang Fig. 5) and hole concentration p (Liang Fig. 6), recalibrated coupled 60 $\times$ 50: ϕ rises smoothly source $\rightarrow$ drain; holes fill the p-body and are depleted in the n^+ S/D.

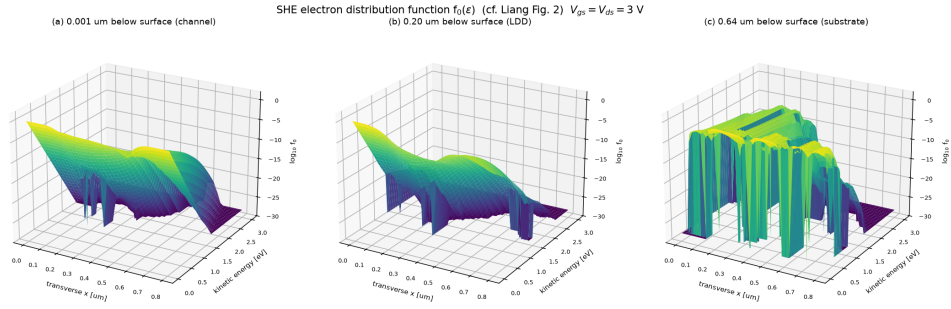


Figure 7: Energy distribution (Liang Fig. 2): a hot, non-Maxwellian tail localized at the drain surface, thermal in the bulk.

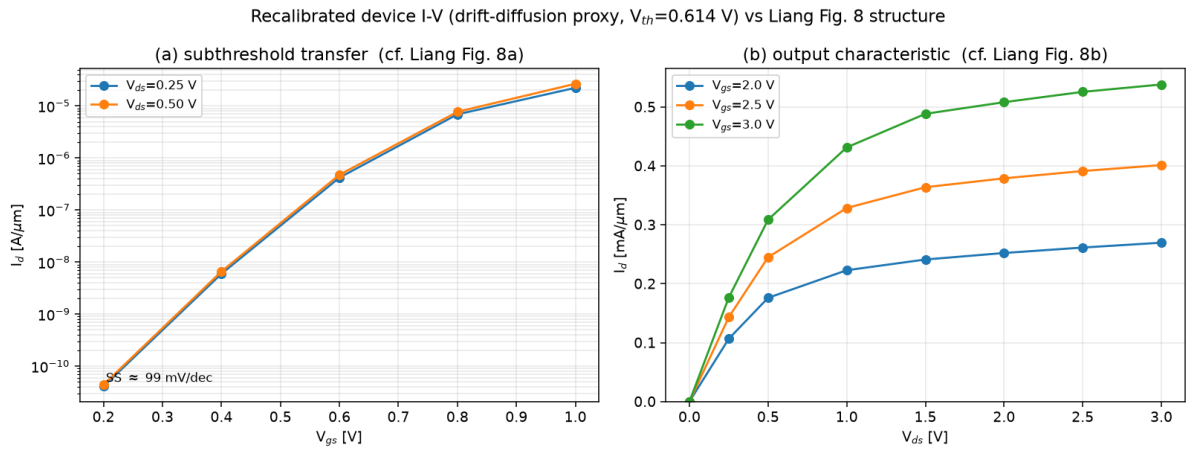


Figure 8: Terminal I-V (Liang Fig. 8), drift-diffusion proxy, per micron: (a) subthreshold transfer at $V_{ds} = 0.25, 0.5$ V; (b) output characteristic at $V_{gs} = 2.0, 2.5, 3.0$ V.

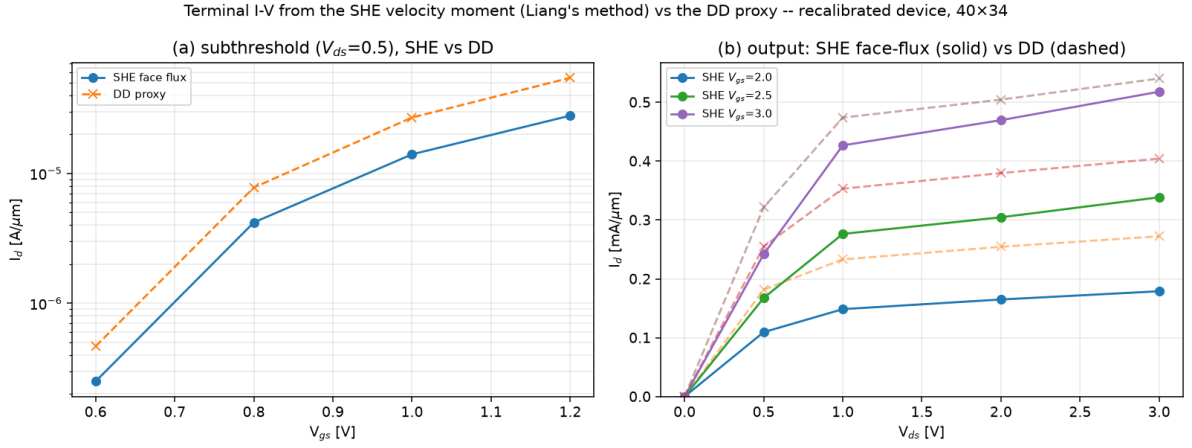


Figure 9: Terminal I-V from the *SHE velocity moment* (finite-volume face-flux current — Liang’s method), one-way SHE on the DD potential, vs the DD proxy (dashed); recalibrated device, 40×34 . (a) subthreshold at $V_{ds} = 0.5$ V *only* — a partial reproduction of Liang’s Fig. 8(a), which also shows 0.25 V; (b) output families $V_{gs} = 2.0, 2.5, 3.0$ V. Source/drain current conservation $|I_{src} + I_{drn}|/|I|$ is $2 \times 10^{-5} - 10^{-3}$ across the sweep (largest at $V_{ds} = 3$ V), so the reported current is the genuine SHE terminal flux. The *measured* ratio $I_{d,SHE}/I_{d,DD}$ is 0.51–0.54 (subthreshold) and 0.60–0.96 (output, rising toward unity at high V_{gs}/V_{ds}): the SHE current is systematically below the mobility-based DD proxy, consistent with the stronger nonlocal/high-field transport in the SHE solution (not attributed to a single mechanism absent a controlled diagnostic).

Per-plot comparison tables (columns: feature / this work / Liang / assessment).

Table 6: Liang Fig. 2 — carrier energy distribution / hot electrons (this work: Fig. 7).

Feature	This work	Liang	Assessment
Drain-surface T_e (tail temp.)	3300–3600 K	hot, non-Maxwellian tail	hot tail reproduced
Source / bulk T_e	300–330 K	thermal	thermal, matches
High-energy tail cutoff	$\varepsilon_{max} = 3.02$ eV (absorbing)	tail to high ε	bounded tail
Localization	drain-surface hot spot	near-drain surface	match

Table 7: Liang Fig. 3 — electron concentration $n(x, y)$ (this work: Fig. 4).

Feature	This work	Liang	Assessment
n_{max} (S/D)	$1.02 \times 10^{20} \text{ cm}^{-3}$ @ (0.10, 0.02)	$\sim 10^{20}$	match
Channel inversion (surface)	$3.1 \times 10^{19} \text{ cm}^{-3}$	inversion layer	match
Drain pinch-off	n decreases toward drain	pinch-off	match

Table 8: Liang Fig. 4 — average electron velocity (this work: Fig. 4).

Feature	This work	Liang	Assessment
$ v _{max}$	$9.9 \times 10^6 \text{ cm/s}$ @ drain	$\sim 10^7$ (sat.)	match
Field pattern	lateral in channel, downward into drain	source→drain, into drain	match
Saturation	velocity-saturated near drain	yes	match

Honesty note. The comparisons apply *our* model on Liang’s *levels*; they are not a pixel registration. With a reconstructed doping profile the absolute spatial positions of the hotspots need not coincide with Liang’s, so structural agreement (drain-localization, T_e broader than G_{ii} , order-of-magnitude levels) is the claim, not point-by-point overlap. $G_{ii,max}$ is unconverged in *both* the mesh and the nonlinear coupling loop (§9), so its 27%-of-top value is a still-falling snapshot, not a bound; and the T_e/G_{ii} peak offset stays within one cell. **The Fig. 8 I-V now includes a SHE velocity-moment current** (Fig. 9): the terminal current is read from the SHE finite-volume face flux, reproducing Liang’s method (current from the BTE moment, *not* a mobility model), with source/drain conservation 2×10^{-5} – 10^{-3} . It is *one-way* (SHE on the DD potential, not the full coupled loop), so it demonstrates the method but is not yet a self-consistent I-V. The DD proxy is retained for comparison; the measured $I_{d,SHE}/I_{d,DD} = 0.5$ – 0.96 (bias-dependent), systematically below DD — consistent with the stronger nonlocal/high-field transport in the SHE, though not attributed to a single mechanism here. Liang’s Fig. 8 solid lines are moreover *experimental* data for a real device of unstated width, so only the per-width shape is comparable, not the absolute ampere magnitude.

Table 9: Liang Fig. 5 — electrostatic potential $\phi(x, y)$ (this work: Fig. 6).

Feature	This work	Liang	Assessment
ϕ swing	$-0.35 \rightarrow +3.59$ V (3.93 V)	$V_{ds}=3$ + built-ins	match
Source-surface ϕ	0.59 V (built-in)	built-in	match
Drain-surface ϕ	3.59 V ($= V_{ds}$ +built-in)	$\approx V_{ds}$	match
Profile	smooth, monotone S→D	smooth 2-D	structural (reconstr. doping)

Table 10: Liang Fig. 6 — hole concentration $p(x, y)$ (this work: Fig. 6).

Feature	This work	Liang	Assessment
$p_{\max}$	$2.1 \times 10^{16} \text{ cm}^{-3}$ (p-body)	p-body / substrate	match (structural)
n^+ S/D holes	$\leq 4 \times 10^{13} \text{ cm}^{-3}$ (depleted)	hole-depleted	match
II holes near drain	elevated p from G_{ii} source	avalanche holes	present
Caveat	body-contact BC + reconstr. doping	—	structural, not point-wise

Part III

Assessment

11 Conclusions, classified

- **Scope — reduced reproduction.** This reproduces the principal 2-D *signatures* (transport moments, hot-electron localization, avalanche feedback) with a P1/($\ell \leq 1$) SHE, but is *not* a full reproduction of Liang’s simulator: the explicit surface-roughness, surface-acoustic, and ionized-impurity operators are lumped into τ_1 (Table 2), the band is a 1-parameter Kane surrogate, and the doping is reconstructed. Defensible label: a *reduced P1/SHE-Boltzmann reproduction of the principal signatures*, using reconstructed device data and surrogate scattering/band models.
- **Calibration vs. validation.** The overlay device is calibrated (V_{th} fitted to 0.614 V via Φ_{gate} ; Table 3); quantitative agreements are calibrated comparisons, with $|v|_{\max}$, the T_e/G_{ii} *localization*, and subthreshold swing as the independent checks.
- **Core SHE transport** shows a clean grid-sensitivity hierarchy over the tested $40 \times 34 \rightarrow 60 \times 50$ refinement: n *stable* (0.4%), $|v|$ *stable* (2.7%), T_e moderately sensitive (5.9%; stable, not converged), G_{ii} *unconverged/mesh-dependent* (40%). “Stable over the tested refinement” is a two-grid statement, not a proof of asymptotic convergence.
- **Impact ionization:** now part of the assembled electron collision operator, with explicit primary depletion and secondary redistribution (G_{ii} is a moment of the II-affected f_0); the equal-energy secondary split and negligible scattering-in shape change follow the paper.
- **Self-consistency:** the SHE $\leftrightarrow$ Poisson $\leftrightarrow$ hole feedback is implemented and materially suppresses avalanche generation and drain current, but a 50-iteration study (§9, Fig. 3) shows it reaches only $\max |\Delta\phi| = 2.5 \text{ mV}$ (not the 2 mV criterion) and that G_{ii} *does not converge in the loop* (monotonic decrease, no plateau); n and $|v|$ are stable while T_e drifts down slowly. The previously-quoted -37% feedback fraction is *superseded* — the suppression is $\geq -53\%$ and still growing.

Table 11: Liang Fig. 7 — T_e (dashed) and G_{ii} (solid) contours (this work: Figs. 4,5).

Feature	This work	Liang	Assessment
$T_{e,\max}$	3323 K (66% of top)	to 5000 K	same order, drain-localized
$G_{ii,\max}$	6.3×10^{26} (27% of top)	to 2.3×10^{27}	unconverged in mesh <i>and</i> loop (still falling)
T_e vs G_{ii} extent	T_e region $\sim 30\times$ larger	T_e broader	match
Peak offset	≤ 1 cell (14 nm)	offset	unresolved at mesh

Table 12: Liang Fig. 8 — terminal I-V (this work: Figs. 8,9).

Feature	This work	Liang	Assessment
I-V method	SHE velocity moment (face flux), one-way (Fig. 9)	BTE velocity moment	<i>method reproduced</i> (one-way)
Output shape	triode $\rightarrow$ sat, V_{gs} -ordered (SHE and DD)	Fig. 8b	shape match
$I_{d,\text{SHE}}(V_g=V_d=3)$	0.52 mA/ μm (SHE face flux)	—	from BTE flux, not DD
$I_{d,\text{SHE}}/I_{d,\text{DD}}$	0.51–0.54 (sub.), 0.60–0.96 (out.)	—	SHE below DD (measured)
S/D conservation	2×10^{-5} – 10^{-3}	—	genuine terminal flux
Subthreshold swing	~ 99 mV/dec (DD)	sub- μm NMOS	reasonable

- **Quantitative overlays (this revision):** on Liang’s own Fig. 7 contour levels the best-converged coupled fields reach 66% (T_e) and 27% (G_{ii} , unconverged) of the top published contours, drain-localized and correctly structured (T_e region much broader than G_{ii}); a one-way SHE velocity-moment I-V reproduces Liang’s current-extraction method (Fig. 9), with the DD curve retained for comparison. A per-plot comparison table is given for each of Liang’s Figs. 2–8 (§10, Tables 6–12).
- **Conservation:** validated across one-way, refined, and coupled calculations (number balance 10^{-5} – 10^{-7} ; terminal currents balance $< 0.2\%$).
- **Numerics:** physics-split preconditioning removes the previous practical solver bottleneck.
- G_{ii} : the earlier 1.37×10^{27} agreement with Liang was *demonstrably accidental* — spatial refinement and self-consistency each push it downward (multiplicatively separable to $\sim 1\%$ among the under-converged snapshots). On the original device the coupled 60×50 run gives $5.24 \times 10^{26} \text{ cm}^{-3}\text{s}^{-1}$, the *finest-tested coupled value* (not a continuum estimate: still -38% below coupled 40×34 , no plateau); a one-way energy refinement moves it only -2.4% . *No converged magnitude can be quoted* (Table 5). The 50-iteration study reinforces this: the coupled loop *itself* does not converge G_{ii} (monotonic decrease, Fig. 3), so all coupled entries — Table 5 and the recalibrated 1.34×10^{27} snapshot — are under-converged.
- T_e/G_{ii} **offset:** unresolved at present mesh resolution; *no longer claimed reproduced*.
- **Device:** recalibrated to $V_{th} = 0.614 \text{ V}$ ($\Phi_{gate} = -0.74 \text{ V}$), replacing the earlier over-high 1.64 V.
- **Done this revision:** a one-way *SHE velocity-moment* terminal I-V (Fig. 9, Table 12) — terminal current from the SHE face flux, as in Liang, with source/drain conservation $\leq 10^{-3}$ — upgrading Fig. 8 from a DD shape check to a demonstrated (if not yet self-consistent) reproduction of Liang’s I-V method.

- **Open items, in priority order.** (a) an *accelerated* coupled solver (Anderson/Newton): the damped Picard loop reaches only 2.5 mV in 50 iterations and does not converge G_{ii} (§9), which is now the *fundamental* G_{ii} blocker — spatial refinement on an unconverged nonlinear state is hard to interpret, so the fixed point must be controlled first; (b) *then* a third spatial grid to assess G_{ii} spatial convergence after the nonlinear fixed point is controlled (a third grid does not itself guarantee convergence), plus ε_{max} /coupled- ΔH sensitivity; (c) a fully self-consistent (not one-way) SHE-moment I-V; (d) explicit surface-roughness / -acoustic / impurity operators replacing the lumped τ_1 .

A Reproducibility package

All values below are the literal code constants (`src/device.py`, `constants.py`, `she2d.py`, `dd.py`); the simulation is fully specified by them plus the run commands.

Constants. $T = 300$ K; $\varepsilon_{Si} = 11.7$, $\varepsilon_{ox} = 3.9$; $n_i = 1.45 \times 10^{10} \text{ cm}^{-3}$; $m^* = 0.32 m_0$; $g_v = 6$; $\alpha = 0.5 \text{ eV}^{-1}$. Table I: $D_{ac} = 4.0 \text{ eV}$, $D_n = 5.0 \times 10^8 \text{ eV/cm}$, $E_{op} = 0.05 \text{ eV}$, $v_s = 9.0 \times 10^6 \text{ cm/s}$, $\Gamma = 1.96 \text{ cm}^2/\text{V}^2/\text{s}$, $n_{scr} = 1.45 \times 10^{15} \text{ cm}^{-3}$, $D_{sa} = 17.8 \text{ eV}$, $\delta = 13.1$, $\varepsilon_{max} = 3.02 \text{ eV}$, $\rho_{Si} = 2.329 \text{ g/cm}^3$, $u = 9.04 \times 10^5 \text{ cm/s}$. Some paper constants (D_{sa} , Γ , δ , n_{scr}) are retained here for reference but are *not active as distinct collision operators* in the reduced model; Table 2 is authoritative for active model usage.

Geometry (μm). $L_x = 0.80$, $L_y = 0.70$; $t_{ox} = 0.0096$; gate $[0.20, 0.60]$ ($L_{\text{eff}} \approx 0.35$); n^+ edges $x < 0.150$ / $x > 0.650$; LDD edges 0.220 / 0.580 ; $x_{j,n^+} = 0.105$ (metallurgical $0.105 + 2.63\sigma_{dep} = 0.171$), $x_{j,LDD} = 0.060$; straggles $\sigma_{lat} = 0.028$, $\sigma_{dep} = 0.025$; channel implant span $[0.18, 0.62]$, $x_{j,ch} = 0.060$, $\sigma_{ch} = 0.035$.

Doping [cm^{-3}]. $N_D^+ = 10^{20}$, $N_D^{\text{LDD}} = 10^{18}$, $N_{A,\text{sub}} = 10^{16}$, $N_{A,\text{chimp}} = 4 \times 10^{17}$. Analytic erfc profile: lateral $L(x) = \frac{1}{2} \text{erfc}((x-x_e)/\sigma_{lat})$ (source) or its mirror (drain), depth $D(y) = \frac{1}{2} \text{erfc}((y-x_j)/\sigma_{dep})$; $N_D = \sum_k N_D^{(k)} L^{(k)} D^{(k)}$; $N_A = N_{A,\text{sub}} + N_{A,\text{chimp}} \frac{1}{2} [\text{erfc}(\frac{x-x_{c2}}{\sigma_{lat}}) - \text{erfc}(\frac{x-x_{c1}}{\sigma_{lat}})] D_{ch}(y)$; $N_{net} = N_D - N_A$.

Mesh & energy grid. $x = \text{linspace}(0, L_x, N_x)$; surface-refined $y = L_y s^{1.8}$, $s = \text{linspace}(0, 1, N_y)$. SHE grids 40×34 , 60×50 ; DD-I-V $56 \times 48/44 \times 36$. $\Delta H = 12.5 \text{ meV}$ ($E_{op}/\Delta H = 4$); global $-q\phi_{max} \leq H \leq \varepsilon_{max} - q\phi_{min}$; mask $0 \leq H + q\phi \leq \varepsilon_{max}$.

Boundary conditions. Poisson: Dirichlet (built-in+applied) at ohmic S/D/body; gate oxide Robin (ε_{ox}/t_{ox}); Neumann on free edges. SHE: contact Maxwellian $f_0 = n_c e^{-\varepsilon/k_B T} / \int Z e^{-\varepsilon/k_B T} dH$; $\varepsilon=0$ reflecting; ε_{max} absorbing (Dirichlet-to-zero on spatial+acoustic+optical fluxes). Holes: ohmic Dirichlet p_{eq} , G_{ii} pair source.

SHE / scattering. $c_{op} = 4.69 \times 10^{-34} \text{ J m}^3/\text{s}$; $\tau_1 = 10^{-14} \text{ s}/\sqrt{\gamma\gamma'}$; $\tau_{ac} = \tau_1$; impact ionization $1/\tau_{ii} = P[(\varepsilon - \varepsilon_{th})/\varepsilon_{th}]^2 \Theta(\varepsilon - \varepsilon_{th})$, $P = 2 \times 10^{13} \text{ s}^{-1}$, $\varepsilon_{th} = 1.1 \text{ eV}$, two equal-energy secondaries at $(\varepsilon - \varepsilon_{th})/2$. ε_{max} sensitivity: bulk $T_e(300 \text{ kV/cm})$ is invariant to 0.1% over $\varepsilon_{max} \in \{3.02, 4, 5\} \text{ eV}$ (device ε_{max} -refinement is an open coupled study).

Hole / DD proxy. Caughey–Thomas (n : $\mu_{min}=68.5$, $\mu_{max}=1414$, $N_{ref}=9.2 \times 10^{16}$, $a=0.711$; p : $44.9/470.5/2.23 \times 10^{17}/0.719 \text{ cm}^2/\text{Vs}$), field saturation ($v_{sat,n}=1.07 \times 10^7$, $\beta_n=2$; $v_{sat,p}=8.37 \times 10^6$, $\beta_p=1$), SRH $\tau_n=\tau_p=10^{-7} \text{ s}$, SG continuity, Gummel loop with optional bias-continuation warm start; drain current = SG electron flux into drain-contact nodes, per- μm .

Solvers, norms, clipping. SHE: physics-split $E \rightarrow S \rightarrow E$ preconditioner, LGMRES tol 10^{-10} (audit) / 10^{-8} (coupled). Coupled loop: ϕ -damping 0.25–0.35, 2 mV max $|\Delta\phi|$ target. Norms: relative number balance = (net injection – cutoff losses)/total rate; terminal-current mismatch = $|I_{src} + I_{drn}| / \frac{1}{2}(|I_{src}| + |I_{drn}|)$. Clipping: f_0 negatives set to zero (pre-clip min $\sim -3 \times 10^{-11}$, $\sim 46/328544$ nodes; moment change $< 10^{-4}$).

Software, commit, run commands. Python 3.13.14, NumPy 2.5.1, SciPy 1.18.0; code at the commit in Appendix B. From src/: `python coupled_she.py 60 50 -0.74` (recalibrated coupled solve); `python she2d_liang_overlay.py / she2d_liang_compare.py / she2d_paper_compare_figs.py` (Fig. 7 overlays, comparison plots); `python she2d_iv.py` (DD-proxy Fig. 8); `python she2d_iv_moment.py` (SHE velocity-moment I-V, Fig. 9); `python audit_result.py` (regression audit); `python coupled_converge.py 0.35 50` (coupled loop toward 2 mV).

B Provenance and audit history

Base code commit 0994290; Python 3.13.14, NumPy 2.5.1, SciPy 1.18.0. Key commits: `f016e3c` (locked one-way audit), `7aad998` (60×50 refinement), `65461ad` (physics-split solver), `047da93` (coupled loop), `0994290` (recalibrated coupled 60×50). Four audit rounds fixed: over-claims, missing specs, junction geometry, N_c convergence, the reducible-collision zero-field- T_e bug (acoustic operator), the global H -grid, the contact-Maxwellian normalization (report typo; code correct), boundary/velocity/current extraction, and finally delivered the real impact-ionization operator, the coupled driver, and the physics-split solver. Revision 3 adds the quantitative comparison against Liang’s published figures (§10): the recalibrated coupled fields on Liang’s Fig. 7 contour levels, plus potential/hole (Figs. 5/6) and terminal I-V (Fig. 8) plots (`she2d_liang_compare.py`, `she2d_liang_overlay.py`, `she2d_liang_fig5.6.py`, `she2d_iv.py`), with one comparison table per plot (Tables 6–12). Revision 4 responds to an external audit: corrected the Fig. 7 contour levels (added 4800 K, removed a spurious $2.3 \times 10^{26} G_{ii}$ level) and regenerated the overlays; reframed the work as a *reduced* P1/SHE reproduction with a paper-to-code physics matrix (Table 2) and a calibration-vs-validation matrix (Table 3); added a one-way SHE velocity-moment I-V (Fig. 9); fixed the doping reference ([7]→[25]); and added the reproducibility package (Appendix A). The coupled-loop convergence study is now complete (§9, Fig. 3): 50 iterations reach 2.5 mV (not 2 mV) and reveal that G_{ii} does not converge in the outer loop, superseding the earlier snapshot values (overlays regenerated on the best-converged fields). Three-level spatial refinement and coupled $\varepsilon_{max}/\Delta H$ sensitivity remain deferred; the one-way SHE velocity-moment I-V campaign is complete (Fig. 9, Table 12).

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